

123456

Sheet 1

File: io-v2.0-sheet1.kicad_sch

Sheet 2

File: io-v2.0-sheet2.kicad_sch

Sheet 3

File: io-v2.0-sheet3.kicad_sch

Sheet 4

File: io-v2.0-sheet4.kicad_sch

Sheet 5

File: io-v2.0-sheet5.kicad_sch

Sheet 6

File: io-v2.0-sheet6.kicad_sch

Sheet 7

File: io-v2.0-sheet7.kicad_sch

Circuit source: io_v2_netlist.py; generated by gen_io_v2.py; see README.md
Sheets 1-6 = I/O card v1.1 (unchanged); sheet 7 = CompactFlash on P4/P5 (from the CF card v1.0)

Sheet: /
File: io-v2.0.kicad_sch

Title: YACC1 I/O card v2.0

Size: A4Date: 2026-09-23

Rev: v2.0

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A

B

C

D

1

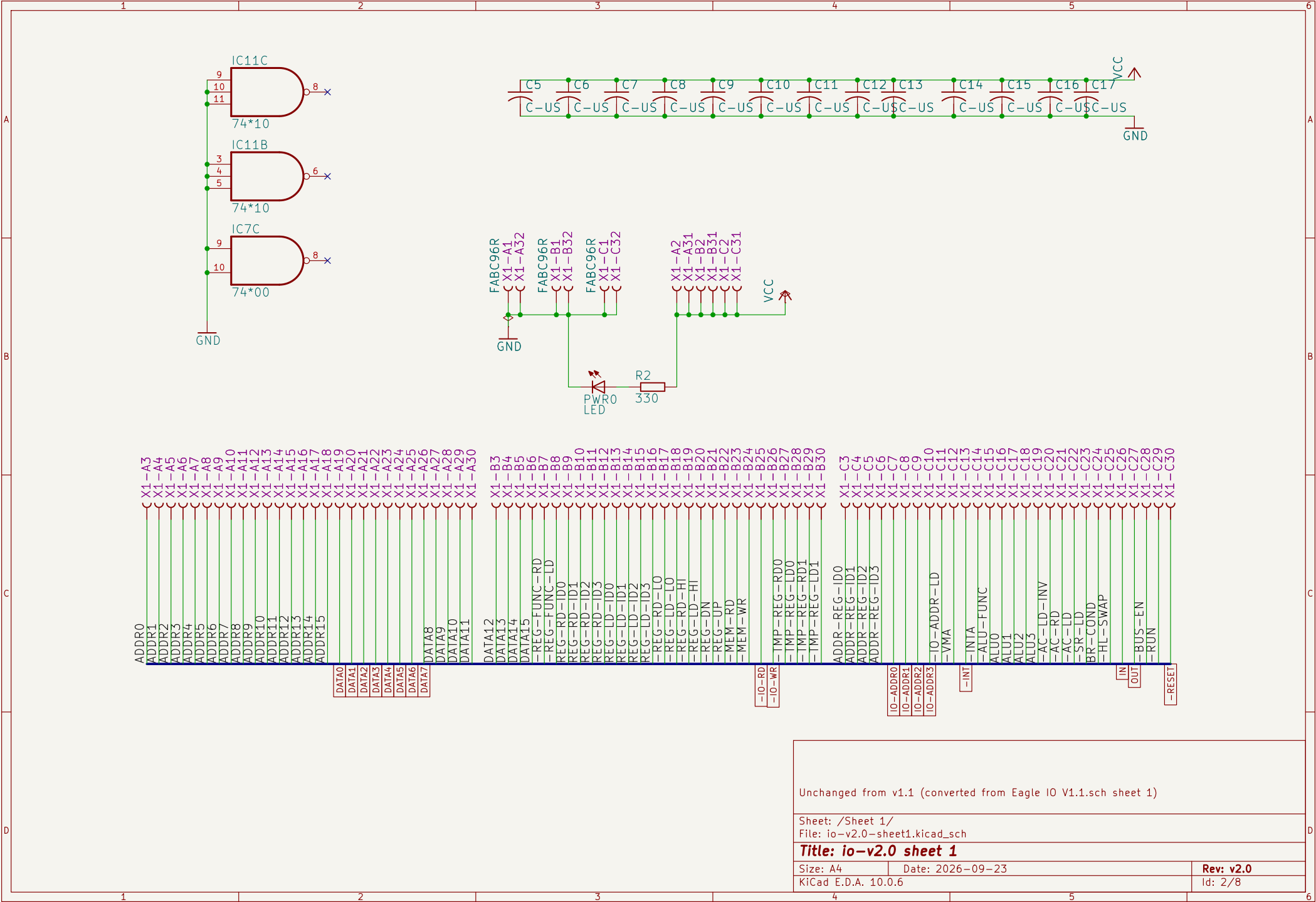
2

3

4

5

6



Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 1)

Sheet: /Sheet 1/
File: io-v2.0-sheet1.kicad_sch

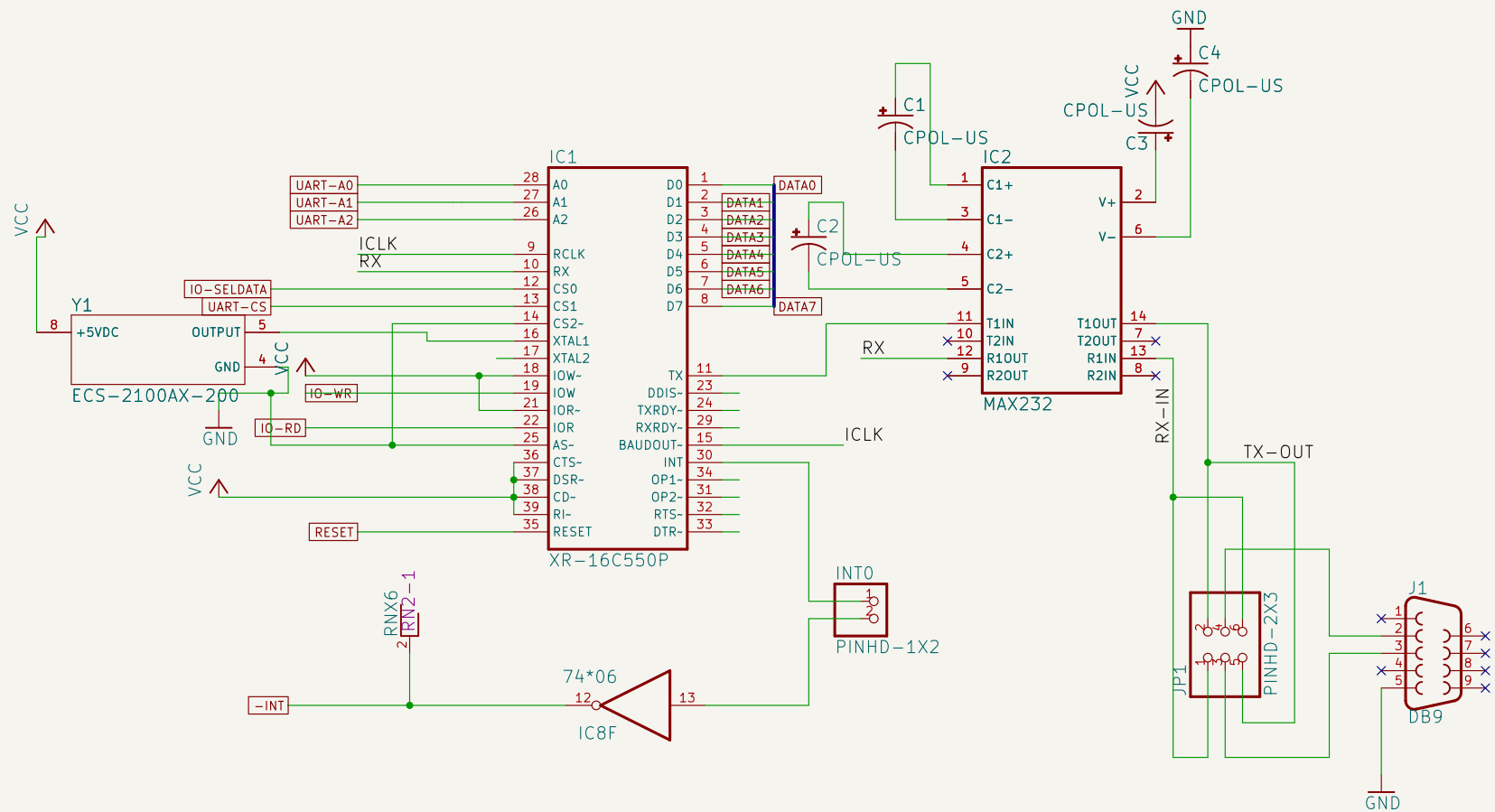
Title: io-v2.0 sheet 1

Size: A4 Date: 2026-09-23

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Rev: v2.0

Id: 2/8



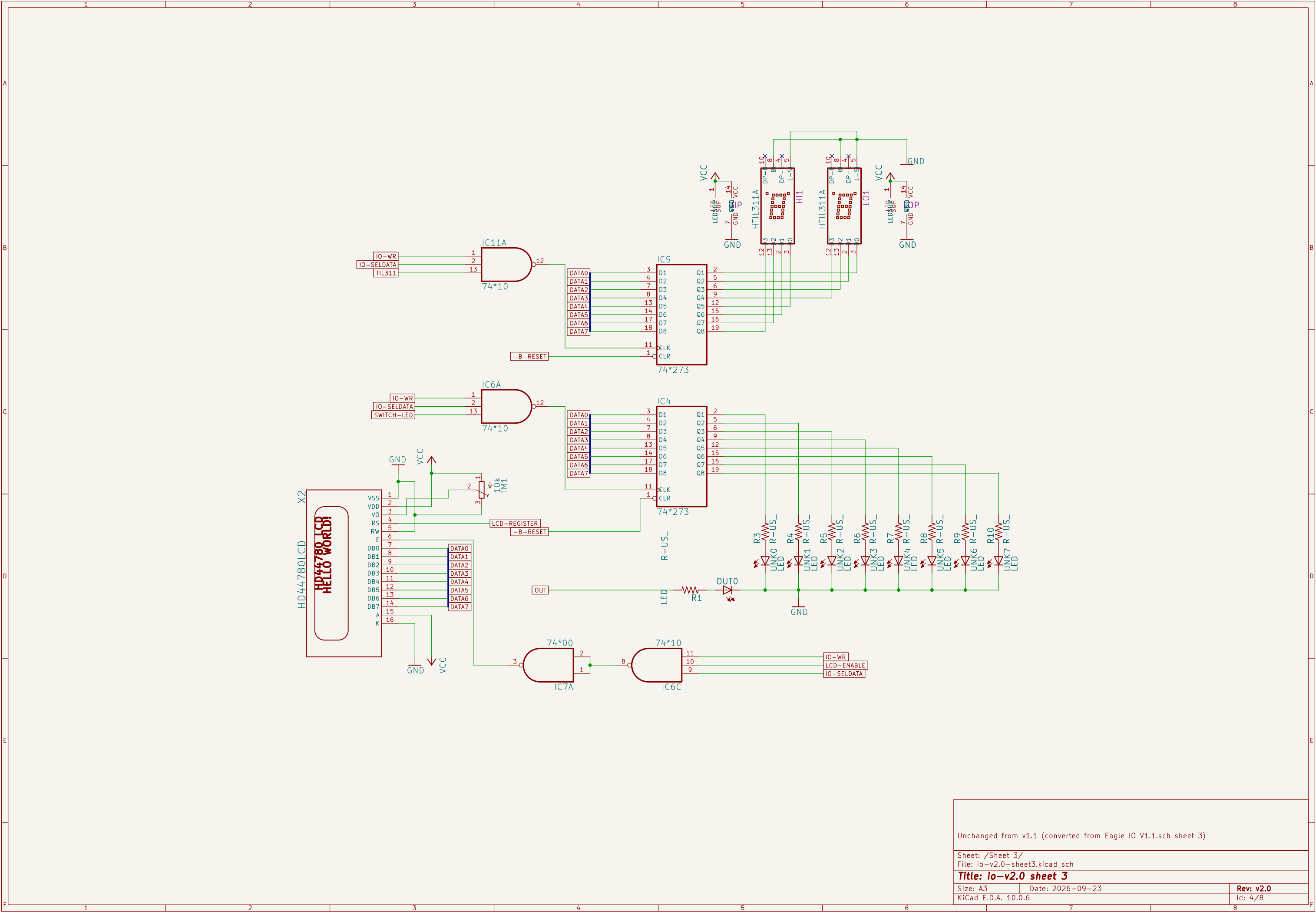
Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 2)

Sheet: /Sheet 2/
File: io-v2.0-sheet2.kicad_sch

Title: io-v2.0 sheet 2

Size: A4 Date: 2026-09-23
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Rev: v2.0
Id: 3/8



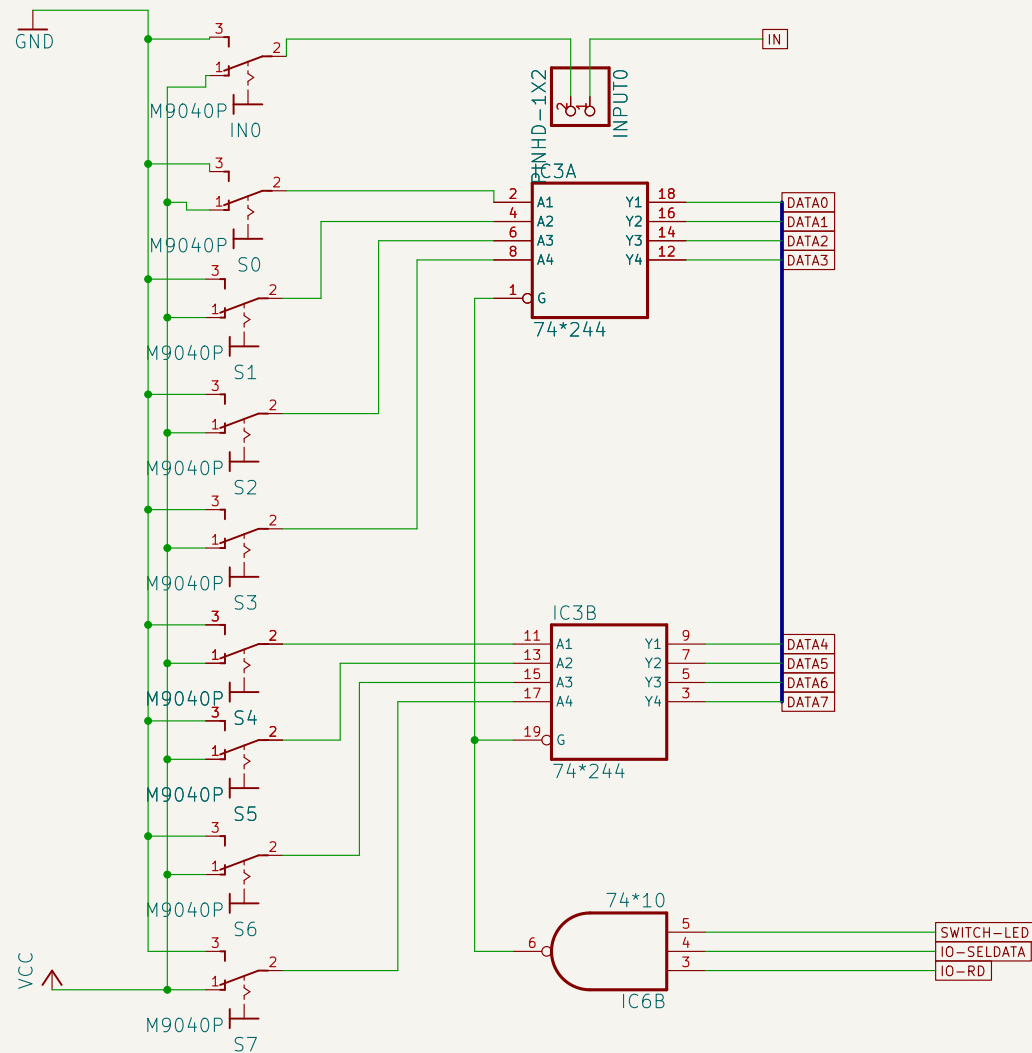
Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 3)

Sheet: /Sheet 3/
File: io-v2.0-sheet3.kicad_sch

Title: io-v2.0 sheet 3

Size: A3
Date: 2026-09-23

Rev: v2.0
Id: 4/8



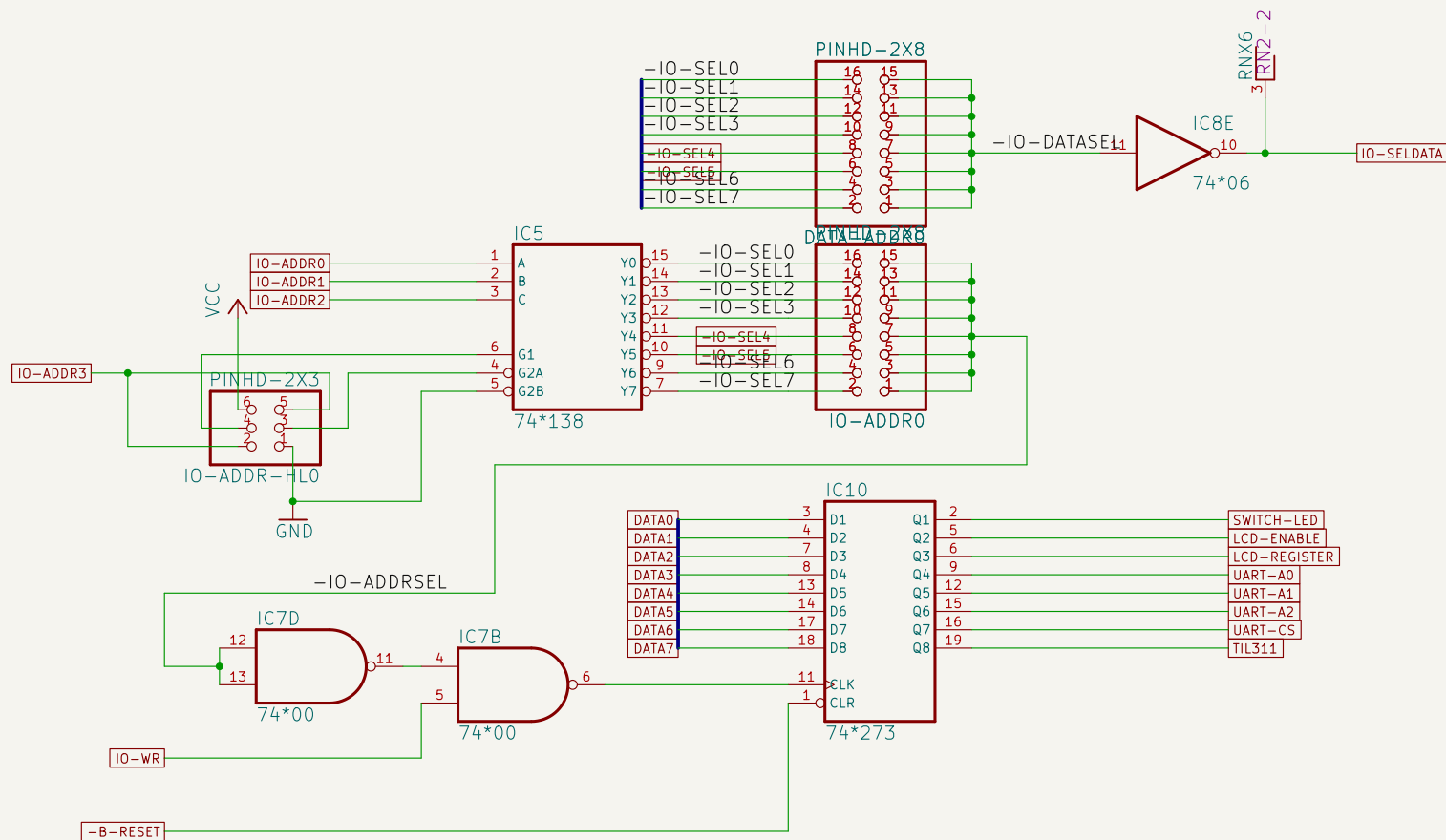
Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 4)

Sheet: /Sheet 4/
File: io-v2.0-sheet4.kicad_sch

Title: io-v2.0 sheet 4

Size: A4 Date: 2026-09-23
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Rev: v2.0
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I/O card v2.0: the IO-ADDR-HL strap MUST decode P0-P7 (jumpers 3-5 and 4-6):
 IC5 Y4 (-IO-SEL4) = P4 = CF register-select latch, Y5 (-IO-SEL5) = P5 = CF data (sheet 7).
 The IO-ADDR and DATA-ADDR jumpers must not select P4 or P5 (pins 7-8, 5-6).

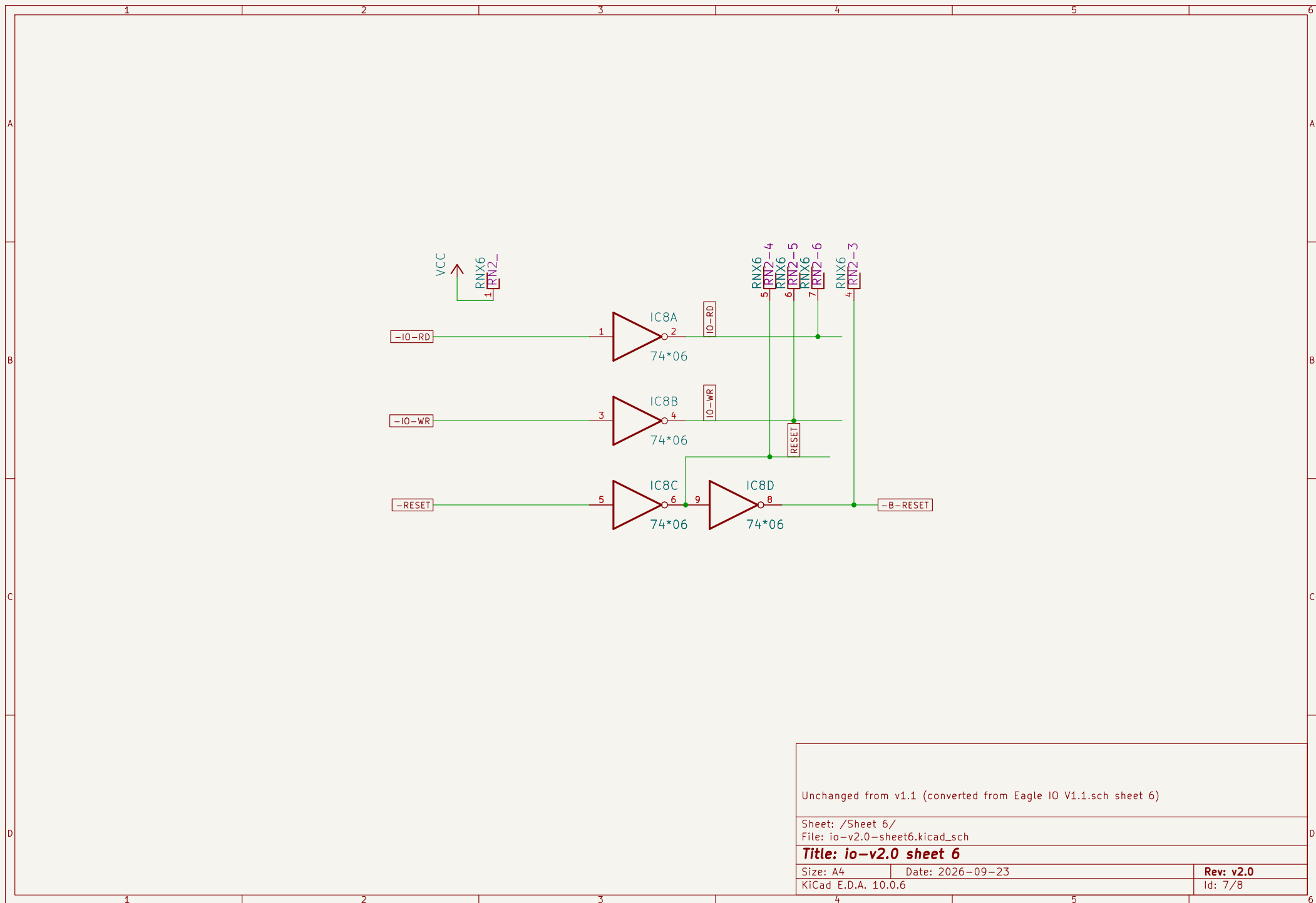
v1.1 sheet 5 + -IO-SEL4/-IO-SEL5 made global (CF section, sheet 7) + strap note

Sheet: /Sheet 5/
 File: io-v2.0-sheet5.kicad_sch

Title: io-v2.0 sheet 5

Size: A4 Date: 2026-09-23
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Rev: v2.0
 Id: 6/8



Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 6)

Sheet: /Sheet 6/
File: io-v2.0-sheet6.kicad_sch

Title: io-v2.0 sheet 6

Size: A4
Date: 2026-09-23

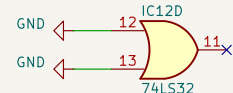
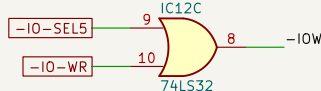
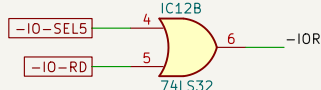
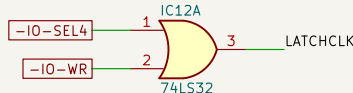
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Rev: v2.0
Id: 7/8

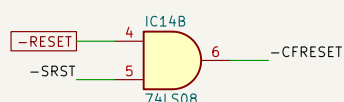
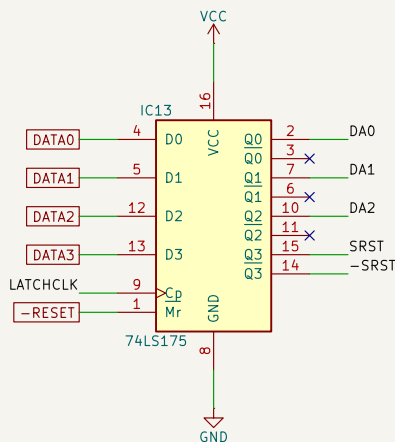
P4 / P5 STROBE GATING (IC12 74LS32)

-IO-SEL4 = IC5 Y4 (sheet 5): port P4
 -IO-SEL5 = IC5 Y5 (sheet 5): port P5
 IC5 decodes P0-P7 only with the
 IO-ADDR-HL strap in P0-P7.

```
gate 1: P4 latch clock
= -IO-SEL4 OR -IO-WR
gate 2: CF -IOR
= -IO-SEL5 OR -IO-RD
gate 3: CF -IOW
= -IO-SEL5 OR -IO-WR
gate 4: unused
```

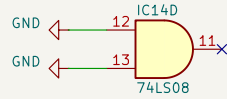
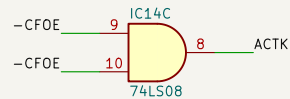
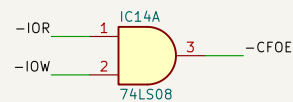
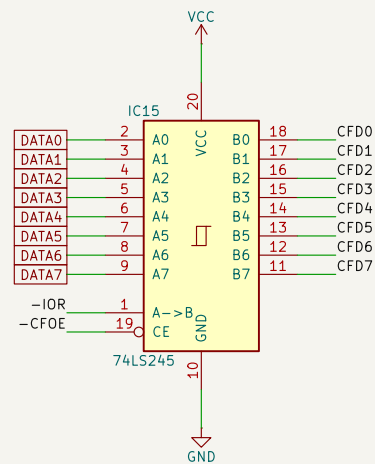


P4 LATCH: DA0-2 + CF RESET (IC13 74LS175)



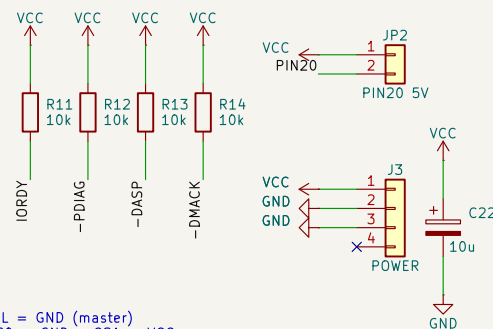
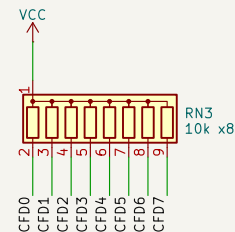
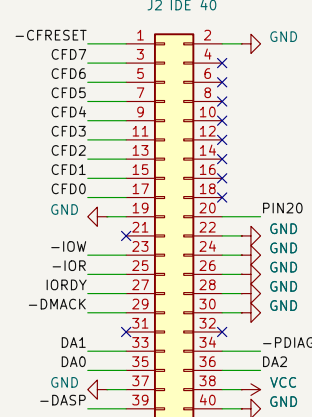
IC14 gate 2: $\text{CFRESET} = \text{RESET AND SRST}$
(bus reset or latch bit 3)

P5 DATA BUFFER (IC15 74LS245)



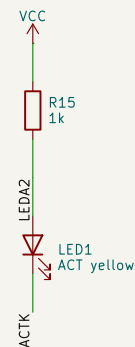
```
DIR = -IOR (low: CF -> bus)
-CFOE = -IOR AND -IOW
ACTK sinks the ACT LED
```

IDE HEADER J2 (TO THE CF ADAPTER)



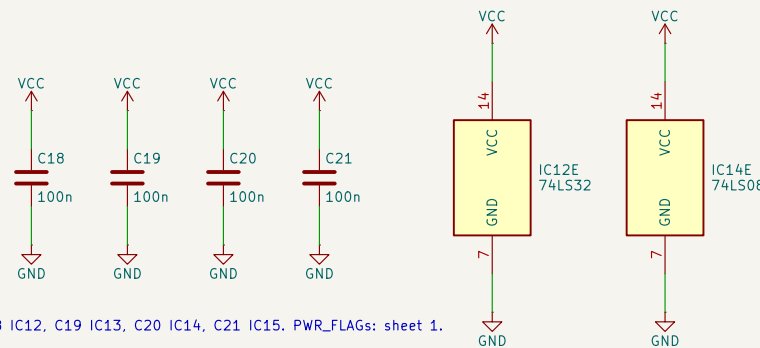
CSEL = GND (master)
-CS0 = GND, -CS1 = VCC

ACT LED



ACT: any P5 access
(PWR LED: the card's PWR0;
no DASP LED on v2.0)

POWER + DECOUPLING (one 100 nF per new IC)



C18 IC12, C19 IC13, C20 IC14, C21 IC15. PWR_FLAGS: sheet 1.

DESIGN NOTES (I/O CARD v2.0, CF SECTION)

- Two I/O ports on the I/O card's own decoder IC5 (sheet 5): P4 (write) = 74LS175 latch IC13 - bits 0-2 = the ATA task-file register (CF DA0-2), bit 3 = CF reset (1 = held in reset); P5 (read/write) = the selected ATA register, 8-bit True IDE, through the 74LS245 IC15. The IO-ADDR-HL strap MUST decode P0-P7; the IO-ADDR / DATA-ADDR jumpers must not select P4 or P5 (they would put the control latch or the UART/switch port on a CF port).
 - IC12 (74LS32) ORs each port select with -IO-WR / -IO-RD, so -IOR, -IOW and the latch clock are low only during the bus strobe for this port (the microcode sets IOADDR two steps before the strobe and holds it through it).
 - CS0 (J2 pin 37) is tied LOW and -CS1 (pin 38) HIGH: the strobes alone define every cycle (CF card v1.0 note 3, docs/cards/cf.md section 4; gating -CS0 would break the chip-select hold time after -IOR).
 - J2 = 40-pin IDE header, a short ribbon to a CF-to-IDE adapter mounted above the card on standoffs. Adapter power: J3 (1 = +5 V, 2 and 3 = GND, 4 = +12 V, not used) or, with JP2 fitted, IDE pin 20 for adapters powered there.
 - Pull-ups: CF D0-7 (RN3; an empty adapter reads \$FF = 'no card' to the ROM driver), IORDY, -PDIAG, -DASP; -DMACK held inactive; CSEL grounded (master). SRST (IC13 Q4) is a one-pin net by design (a probe point).
 - From hardware/cards/cf/kicad/v1.0/cf.netlist.py: U3->IC12 U3->IC13 U4->IC14 U5->IC15 J1->J2 J2->J3 JP1->JP2 RN1->RN3 R1->R5->R11-R15 LED2->LED1 C2->C6->C18-C22. Dropped: U1 (74LS138: IC5 Y4/Y5 replace -P8SEL/-P9SEL), X1 (the card's bus connector), C1, the PWR LED (LED1, R7: the card has PWR0) and the DASP LED (LED3, R6).
- Theory: docs/cards/cf.md, docs/cards/io.md. Circuit source: io_v2.netlist.py (edit, then re-run build.sh).

Circuit source: [io_v2_netlist.py](#); this sheet is generated by [gen_io_v2.py](#)
 From the CF card v1.0 minus its 74LS138, bus connector, PWR + DASP LEDs
 CF (True IDE, 8-bit) on I/O ports P4 (register latch) / P5 (data) – [docs/cards/cf.md](#)
YACC1

Sheet: /Sheet 7/
File: io-v2.0-sheet7.kicad_sch

Title: io-v2.0 sheet 7: CompactFlash interface

Size: A3	Date: 2026-09-23
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